

1(a)	any sensible concentrations ; corresponding volumes of distilled water <u>and</u> sucrose;	2									
1(b)	any seven from: 1 same / stated, variety / age / AW, of beetroots ; 2 remove the outer skin (of the beetroot) / AW ; 3 same / stated, dimensions / AW, of (beetroot) block / AW ; 4 <i>ref. to</i> apparatus for obtaining same size of (beetroot) blocks ; 5 for each sucrose concentration, place (beetroot) block in a suitable container <u>and</u> submerge / AW (in sucrose solution); 6 (for each sucrose concentration) measure the initial mass <u>and</u> the final mass ; 7 (measure final mass) after a (suitable) stated time ; 8 method to remove excess liquid from beetroot (before measuring final mass) ; 9 use at least three different (beetroot) blocks <u>and</u> calculate a mean, for each sucrose concentration ; 10 safety comment with named hazard <u>and</u> risk <u>and</u> precaution ; <table border="1"> <thead> <tr> <th>hazard</th><th>risk</th><th>precaution</th></tr> </thead> <tbody> <tr> <td>beetroot</td><td>allergy / irritant</td><td>gloves / mask / PPE</td></tr> <tr> <td>knife / scalpel / cork borer / AW</td><td>cut</td><td>cut away from the hand / cut onto a, board / tile / AW</td></tr> </tbody> </table>	hazard	risk	precaution	beetroot	allergy / irritant	gloves / mask / PPE	knife / scalpel / cork borer / AW	cut	cut away from the hand / cut onto a, board / tile / AW	7
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1(c)(i)	1 appropriate axis labels <u>and</u> units ; 2 sketch shows decreasing line crossing x-axis ;	2									

Question	Answer	Marks
1(c)(ii)	(identify the) x -axis intercept point or (identify the point on the x -axis where the mean) percentage change in mass is zero ;	1
1(c)(iii)	rinse (beetroot) blocks (after cutting / before adding to sucrose solution) ;	1

Question	Answer	Marks
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